

Modelling Large Spatial Data: An Overview and a Deep Dive into Hierarchical Nearest-Neighbor Gaussian Process (NNGP) Models

Shamriddha De and Shagun Srivastava

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1 Introduction

Spatial data arise when responses are observed at geographical locations, and a defining feature of such data is spatial dependence. For example, in a real estate domain, transaction values may be recorded together with covariates such as floor area, number of bedrooms and bathrooms, and other amenities. Transactions located close to one another are generally expected to exhibit greater dependence than geographically distant transactions. A natural way to account for this dependence is through a Gaussian process (GP) spatial regression model.

1.1 Full Gaussian Process Spatial Regression Model

Suppose we observe the data at N locations $\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_N$. A full GP spatial regression model can be expressed as

$$Y(\mathbf{s}_i) = \boldsymbol{\beta}^\top \mathbf{x}(\mathbf{s}_i) + w(\mathbf{s}_i) + \varepsilon(\mathbf{s}_i), \quad i = 1, 2, \dots, N, \quad (1)$$

where, corresponding to the i th observation, $Y(\mathbf{s}_i)$ is the response, $\mathbf{x}(\mathbf{s}_i)$ is the $p \times 1$ vector of covariates, $\boldsymbol{\beta}$ is the $p \times 1$ vector of regression coefficients, $w(\mathbf{s}_i)$ is the spatial random effect arising from a GP, and $\varepsilon(\mathbf{s}_i)$ is random noise uncorrelated with $w(\mathbf{s}_i)$. In matrix notation, the aforesaid model becomes

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{w} + \boldsymbol{\varepsilon}, \quad (2)$$

where $\mathbf{Y} = (Y(\mathbf{s}_1), Y(\mathbf{s}_2), \dots, Y(\mathbf{s}_N))^\top$ is the response vector, $\mathbf{X} = (\mathbf{x}(\mathbf{s}_1)^\top, \mathbf{x}(\mathbf{s}_2)^\top, \dots, \mathbf{x}(\mathbf{s}_N)^\top)^\top$ is the design matrix, $\mathbf{w} = (w(\mathbf{s}_1), w(\mathbf{s}_2), \dots, w(\mathbf{s}_N))^\top$ is the $N \times 1$ vector of spatial random effects and $\boldsymbol{\varepsilon} = (\varepsilon(\mathbf{s}_1), \varepsilon(\mathbf{s}_2), \dots, \varepsilon(\mathbf{s}_N))^\top$ denotes the random noise. We assume that $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Omega})$ and $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \tau^2 \mathbf{I}_N)$, and $\mathbf{w}$ and $\boldsymbol{\varepsilon}$ are uncorrelated. Accordingly, the likelihood becomes

$$\mathbf{Y} \sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta}, \boldsymbol{\Sigma}), \quad \boldsymbol{\Sigma} = \boldsymbol{\Omega} + \tau^2 \mathbf{I}_N. \quad (3)$$

It is to be noted that the spatial dependence among the observations typically results $\boldsymbol{\Omega}$ (and accordingly $\boldsymbol{\Sigma}$) to be a full rank non-diagonal matrix. Furthermore, the likelihood computation

involves $|\Sigma|$ and Σ^{-1} , which require $O(N^3)$ flops and a storage of $O(N^2)$. For large values of N , both the computation time and storage become formidable. The present tutorial discusses a few possible remedies to tackle this issue.

1.2 Possible Approximations for Large Spatial Data

Two broad strategies can be deployed to alleviate the computational and storage burden associated with large spatial datasets. One of them performs low rank approximation of the spatial covariance matrix Ω by expressing the spatial process using a combination of relatively smaller number of representative spatial effects (called knots). The other class of methods induce sparsity in the covariance/precision structure, while retaining the entire spatial process. That is, the spatial dependence in this case is represented primarily through nearby locations. A few examples of both the classes of techniques are listed below:

- **Low Rank Approximation Techniques:**

- Fixed rank kriging ([Cressie and Johannesson, 2006, 2008](#))
- Lattice kriging ([Nychka et al., 2015](#))
- Multiresolution approximation ([Katzfuss, 2017](#))

- **Sparse Inducing Techniques:**

- Spatial partitioning ([Sang et al., 2011](#); [Anderson et al., 2014](#); [Neelon et al., 2014](#))
- Nearest neighbor Gaussian process ([Datta et al., 2016](#); [Finley et al., 2022](#))
- Covariance tapering ([Kaufman et al., 2008](#))

A more extensive review of these methods can be found in [Heaton et al. \(2019\)](#). In this tutorial, we shall briefly skim over fixed rank kriging and spatial partitioning, and subsequently explore nearest neighbor Gaussian process in greater detail.

2 Spatial Partitioning

In this method, we induce sparsity in the spatial covariance by dividing the spatial domain under study into mutually exclusive and exhaustive regions, which are assumed to be spatially independent among themselves. More specifically, a spatial domain $\mathcal{D}$ is divided into K disjoint regions $\{\mathcal{D}_1, \mathcal{D}_2, \dots, \mathcal{D}_K\}$, such that

$$\mathcal{D} = \mathcal{D}_1 \cup \mathcal{D}_2 \cup \dots \cup \mathcal{D}_K, \quad \mathcal{D}_d \cap \mathcal{D}_{d'} = \emptyset \text{ for } d \neq d'. \quad (4)$$

Accordingly, the spatial regression model (2) can be broken down into region-wise models as

$$\mathbf{Y}_d = \mathbf{X}_d \beta_d + \mathbf{w}_d + \varepsilon_d, \quad d = 1, 2, \dots, K. \quad (5)$$

The assumption of spatially independent regions implies a block-diagonal structure of the spatial covariance matrix $\mathbf{\Omega}$ as

$$\mathbf{\Omega} \approx \mathbf{\Omega}^* = \text{blockdiag} \{ \mathbf{\Omega}_1, \mathbf{\Omega}_2, \dots, \mathbf{\Omega}_K \}, \quad (6)$$

with the region-wise spatial covariances appearing as the block-diagonal entries. This automatically ensures a block-diagonal structure in the covariance matrix of the likelihood as follows:

$$\mathbf{\Sigma} \approx \mathbf{\Sigma}^* = \mathbf{\Omega}^* + \tau^2 \mathbf{I}_N = \text{blockdiag} \{ \mathbf{\Sigma}_1, \mathbf{\Sigma}_2, \dots, \mathbf{\Sigma}_K \}, \quad (7)$$

where $\mathbf{\Sigma}_d = \mathbf{\Omega}_d + \tau^2 \mathbf{I}$, for every $d = 1, 2, \dots, K$. The joint likelihood then factorizes in terms of the region-wise likelihoods as

$$p(\mathbf{Y}) \approx \prod_{d=1}^K p(\mathbf{Y}_d) \quad \implies \quad \log p(\mathbf{Y}) \approx \sum_{d=1}^K \log p(\mathbf{Y}_d). \quad (8)$$

The K summands in the log-likelihood in (8) involve region-wise inversions and determinants of covariance matrices, which can be computed in parallel, thereby enhancing computational and storage efficiency. Under a balanced partition, the computation time and storage are $O(N^3/K^2)$ and $O(N^2/K)$ respectively, as opposed to $O(N^3)$ and $O(N^2)$ for full GP.

A crucial aspect in this method is the partitioning scheme adopted for dividing the entire spatial domain into regions. Examples of such partitions are listed below:

- Based on predefined regions (for instance, macro markets for real estate problems);
- Based on equal areas;
- Based on hierarchical clustering algorithms;
- Based on mixture models.

3 Fixed Rank Kriging

This low rank approximation technique aims at expressing the spatial covariance process as a linear combination of a relatively smaller number of basis functions. In particular, the spatial process is represented as a linear combination of M ($\ll N$) basis functions as

$$w(\mathbf{s}_i) \approx \sum_{m=1}^M \eta_m h_m(\mathbf{s}_i), \quad i = 1, 2, \dots, N, \quad M \ll N, \quad (9)$$

where $h_1(\mathbf{s}_i), h_2(\mathbf{s}_i), \dots, h_M(\mathbf{s}_i)$ are the basis functions, and $\eta_1, \eta_2, \dots, \eta_M$ are the corresponding random basis coefficients. The approximate spatial model then turns out to be

$$\mathbf{Y} \approx \mathbf{Y}^* = \mathbf{X}\boldsymbol{\beta} + \mathbf{H}\boldsymbol{\eta} + \boldsymbol{\epsilon}, \quad (10)$$

where

$$\mathbf{H} = \begin{pmatrix} h_1(\mathbf{s}_1) & h_2(\mathbf{s}_1) & \dots & h_M(\mathbf{s}_1) \\ h_1(\mathbf{s}_2) & h_2(\mathbf{s}_2) & \dots & h_M(\mathbf{s}_2) \\ \vdots & \vdots & \ddots & \vdots \\ h_1(\mathbf{s}_N) & h_2(\mathbf{s}_N) & \dots & h_M(\mathbf{s}_N) \end{pmatrix}, \quad \boldsymbol{\eta} = (\eta_1, \eta_2, \dots, \eta_M)^\top.$$

We note that the covariance matrix of the random basis coefficients

$$\text{Var}(\boldsymbol{\eta}) = \boldsymbol{\Lambda} \quad (11)$$

has rank $M \ll N$. Accordingly, the approximate covariance structure of the likelihood becomes

$$\boldsymbol{\Sigma} \approx \boldsymbol{\Sigma}^* = \mathbf{H}\boldsymbol{\Lambda}\mathbf{H}^\top + \tau^2\mathbf{I}_N. \quad (12)$$

Such an approximation leads to

$$\begin{aligned} |\boldsymbol{\Sigma}^*| &= (\tau^2)^N \left| \mathbf{I}_M + (\tau^2)^{-1} \boldsymbol{\Lambda} \mathbf{H}^\top \mathbf{H} \right| \\ \boldsymbol{\Sigma}^{*-1} &= (\mathbf{H}\boldsymbol{\Lambda}\mathbf{H}^\top + \tau^2\mathbf{I}_N)^{-1} = (\tau^2)^{-1} \left[\mathbf{I}_N - \mathbf{H}(\tau^2\boldsymbol{\Lambda}^{-1} + \mathbf{H}^\top\mathbf{H})^{-1}\mathbf{H}^\top \right], \end{aligned} \quad (13)$$

thereby ensuring that the likelihood computation involves determinants and inverses of only $M \times M$ matrices. The resulting computational complexity and storage are $O(NM^2)$ and $O(NM)$ respectively, indicating a significant gain from a full GP.

As far as choice of basis functions is concerned, some commonly used options are the radial basis functions and the B-splines. Moreover, some plausible choices for M are $M = \sqrt{N}$ and $M = \log N$.

4 Nearest Neighbor Gaussian Process (NNGP)

As discussed earlier, a full Gaussian process provides a natural and flexible framework for modeling spatial dependence, but its computational cost of $O(N^3)$ and storage requirement of $O(N^2)$ make its direct implementation difficult for large spatial datasets. The two approximation strategies discussed above address this problem in different ways. Spatial partitioning induces sparsity by dividing the spatial domain into regions and assuming independence across the regions. Although this substantially reduces computation and allows region-wise parallelization, spatial dependence across the boundaries of the partitions is ignored. Fixed rank kriging, on the other hand, reduces the dimension of the spatial process by representing it through a much smaller number of basis functions. This is computationally attractive, but when the number of basis functions is small, the resulting process can be overly smooth and may fail to capture fine-scale spatial variation. The nearest neighbor Gaussian process (NNGP) proposed by [Datta et al. \(2016\)](#) takes a different approach. Instead of reducing the dimension of the spatial process or dividing the domain into independent regions, it induces sparsity by allowing each location to depend only on a small number of its nearest neighbors. Importantly, this construction defines a valid stochastic process over the

entire spatial domain. Thus, parameter estimation, spatial process inference, and prediction at new locations can all be carried out within the same process-based framework, while retaining the computational advantages of a sparse model.

4.1 NNGP Model Construction

Consider a q -variate spatial process defined over a spatial domain $\mathcal{D} \subseteq \mathbb{R}^d$,

$$\mathbf{w}(\mathbf{s}) \sim GP(\mathbf{0}, \mathbf{C}(\cdot, \cdot | \boldsymbol{\theta})), \quad \mathbf{s} \in \mathcal{D}, \quad (14)$$

where

$$\mathbf{w}(\mathbf{s}) = (w_1(\mathbf{s}), \dots, w_q(\mathbf{s}))^\top \in \mathbb{R}^q. \quad (15)$$

Here, d is the dimension of the spatial domain, q denotes the number of spatial processes, and $\boldsymbol{\theta}$ contains the parameters governing the spatial covariance structure. The corresponding $q \times q$ cross-covariance function is

$$\mathbf{C}(\mathbf{s}, \mathbf{t}) = [\text{Cov}\{w_i(\mathbf{s}), w_j(\mathbf{t})\}]_{q \times q}. \quad (16)$$

For a univariate spatial process, $q = 1$, while for spatial locations represented by longitude and latitude, one typically has $d = 2$.

Let

$$\mathcal{S} = \{\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_k\} \quad (17)$$

be a fixed set of k distinct locations, referred to as the *reference set*. Define

$$\mathbf{w}_{\mathcal{S}} = \left(\mathbf{w}(\mathbf{s}_1)^\top, \dots, \mathbf{w}(\mathbf{s}_k)^\top \right)^\top. \quad (18)$$

Thus, $\mathbf{w}_{\mathcal{S}}$ is a $qk \times 1$ vector and, under the parent Gaussian process,

$$\mathbf{w}_{\mathcal{S}} \sim \mathcal{N}(\mathbf{0}, \mathbf{C}_{\mathcal{S}}(\boldsymbol{\theta})), \quad (19)$$

where $\mathbf{C}_{\mathcal{S}}(\boldsymbol{\theta})$ is a $qk \times qk$ positive-definite block covariance matrix whose (i, j) th block is $\mathbf{C}(\mathbf{s}_i, \mathbf{s}_j)$.

The joint density of the spatial process at the reference locations can always be expressed as a product of successive conditional densities:

$$p(\mathbf{w}_{\mathcal{S}}) = p\{\mathbf{w}(\mathbf{s}_1)\} \prod_{i=2}^k p\{\mathbf{w}(\mathbf{s}_i) \mid \mathbf{w}(\mathbf{s}_1), \dots, \mathbf{w}(\mathbf{s}_{i-1})\}. \quad (20)$$

The difficulty with (20) is that the number of locations in the conditioning set increases with i . Consequently, for locations appearing later in the ordering, the conditional distributions involve increasingly large covariance matrices.

NNGP replaces these large conditioning sets by small nearest-neighbor sets. For each $\mathbf{s}_i$, let

$$N(\mathbf{s}_i) \subseteq \{\mathbf{s}_1, \dots, \mathbf{s}_{i-1}\} \quad (21)$$

contain at most m nearest previous locations, where $m \ll k$. The approximate joint density is then defined as

$$\tilde{p}(\mathbf{w}_S) = \prod_{i=1}^k p\{\mathbf{w}(\mathbf{s}_i) \mid \mathbf{w}_{N(\mathbf{s}_i)}\}, \quad (22)$$

where $\mathbf{w}_{N(\mathbf{s}_i)}$ denotes the spatial process values at the selected neighbors of $\mathbf{s}_i$. Therefore, instead of conditioning on all previous locations, each location is allowed to depend only on at most m nearby locations.

The neighbor structure in (22) can be represented using a directed graph. Each spatial location is treated as a node and, whenever $\mathbf{s}_j \in N(\mathbf{s}_i)$, a directed edge connects $\mathbf{s}_i$ to $\mathbf{s}_j$. Since the neighbors of $\mathbf{s}_i$ are chosen only from the locations preceding $\mathbf{s}_i$ in the ordering, the resulting graph contains no directed cycles. Hence, the neighbor graph is a directed acyclic graph (DAG). This acyclic structure is important because it ensures that the product of conditional densities in (22) defines a proper joint density.

For illustration, a cyclic and an acyclic neighbor structure can be represented as follows:

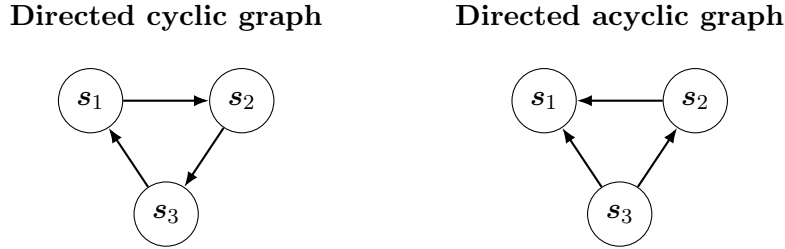


Figure 1: Illustration of directed cyclic and directed acyclic neighbor structures.

Since the parent process is Gaussian, every conditional distribution appearing in (22) is also Gaussian. Accordingly,

$$\tilde{p}(\mathbf{w}_S) = \prod_{i=1}^k \mathcal{N}(\mathbf{w}(\mathbf{s}_i) \mid \mathbf{B}_{\mathbf{s}_i} \mathbf{w}_{N(\mathbf{s}_i)}, \mathbf{F}_{\mathbf{s}_i}), \quad (23)$$

where

$$\mathbf{B}_{\mathbf{s}_i} = \mathbf{C}_{\mathbf{s}_i, N(\mathbf{s}_i)} \mathbf{C}_{N(\mathbf{s}_i)}^{-1}, \quad (24)$$

and

$$\mathbf{F}_{\mathbf{s}_i} = \mathbf{C}(\mathbf{s}_i, \mathbf{s}_i) - \mathbf{C}_{\mathbf{s}_i, N(\mathbf{s}_i)} \mathbf{C}_{N(\mathbf{s}_i)}^{-1} \mathbf{C}_{N(\mathbf{s}_i), \mathbf{s}_i}. \quad (25)$$

Here, $\mathbf{B}_{\mathbf{s}_i} \mathbf{w}_{N(\mathbf{s}_i)}$ is the conditional mean of the spatial process at $\mathbf{s}_i$ given its neighbors, whereas $\mathbf{F}_{\mathbf{s}_i}$ is the corresponding conditional covariance.

The use of small neighbor sets directly induces sparsity in the corresponding precision matrix. In particular, if $N(\mathbf{s}_i)$ contains at most m members for every $\mathbf{s}_i \in \mathcal{S}$, where $m \ll k$, then the

NNGP precision matrix $\tilde{\mathbf{C}}_{\mathcal{S}}^{-1}$ is sparse with at most

$$\frac{k m(m+1)q^2}{2} \quad (26)$$

non-zero entries. Thus, the large dense matrix operations required by a full GP are replaced by computations involving only small neighbor-specific covariance matrices.

So far, the NNGP density has been defined only over the reference set $\mathcal{S}$. Now, consider a location $\mathbf{u} \notin \mathcal{S}$. Let $N(\mathbf{u})$ denote the set of its m nearest neighbors chosen from $\mathcal{S}$. The conditional distribution of the process at $\mathbf{u}$ is defined by

$$\mathbf{w}(\mathbf{u}) \mid \mathbf{w}_{\mathcal{S}} \sim \mathcal{N}(\mathbf{B}_{\mathbf{u}} \mathbf{w}_{N(\mathbf{u})}, \mathbf{F}_{\mathbf{u}}), \quad (27)$$

where

$$\mathbf{B}_{\mathbf{u}} = \mathbf{C}_{\mathbf{u}, N(\mathbf{u})} \mathbf{C}_{N(\mathbf{u})}^{-1}, \quad (28)$$

and

$$\mathbf{F}_{\mathbf{u}} = \mathbf{C}(\mathbf{u}, \mathbf{u}) - \mathbf{C}_{\mathbf{u}, N(\mathbf{u})} \mathbf{C}_{N(\mathbf{u})}^{-1} \mathbf{C}_{N(\mathbf{u}), \mathbf{u}}. \quad (29)$$

Therefore, a new location is connected to the underlying spatial process through only its m nearest reference locations rather than through the entire reference set.

More generally, consider a finite set

$$\mathcal{U} = \{\mathbf{u}_1, \dots, \mathbf{u}_r\}, \quad \mathcal{U} \cap \mathcal{S} = \emptyset. \quad (30)$$

The conditional density of the corresponding spatial process values is defined as

$$\tilde{p}(\mathbf{w}_{\mathcal{U}} \mid \mathbf{w}_{\mathcal{S}}) = \prod_{i=1}^r p(\mathbf{w}(\mathbf{u}_i) \mid \mathbf{w}_{N(\mathbf{u}_i)}). \quad (31)$$

Consequently,

$$\mathbf{w}_{\mathcal{U}} \mid \mathbf{w}_{\mathcal{S}} \sim \mathcal{N}(\mathbf{B}_{\mathcal{U}} \mathbf{w}_{\mathcal{S}}, \mathbf{F}_{\mathcal{U}}), \quad (32)$$

where

$$\mathbf{F}_{\mathcal{U}} = \text{blockdiag}\{\mathbf{F}_{\mathbf{u}_1}, \dots, \mathbf{F}_{\mathbf{u}_r}\}. \quad (33)$$

The matrix $\mathbf{B}_{\mathcal{U}}$ has dimension $r q \times k q$ and is sparse, with each row containing atmost $m q$ non-zero entries.

The above construction allows NNGP to define a valid spatial process over the entire domain. Specifically, for any finite collection of locations

$$\mathcal{V} = \{\mathbf{v}_1, \dots, \mathbf{v}_r\} \subset \mathcal{D}, \quad (34)$$

the resulting process satisfies

$$\mathbf{w}_\mathcal{V} \sim \mathcal{N}(\mathbf{0}, \tilde{\mathbf{C}}_\mathcal{V}), \quad (35)$$

where $\tilde{\mathbf{C}}_\mathcal{V}$ is formed using the NNGP cross-covariance function

$$\tilde{\mathbf{C}}(\mathbf{v}_1, \mathbf{v}_2) = \begin{cases} \tilde{\mathbf{C}}_{\mathbf{s}_i, \mathbf{s}_j}, & \mathbf{v}_1, \mathbf{v}_2 \in \mathcal{S}, \\ \mathbf{B}_{\mathbf{v}_1} \tilde{\mathbf{C}}_{N(\mathbf{v}_1), \mathbf{s}_j}, & \mathbf{v}_1 \notin \mathcal{S}, \quad \mathbf{v}_2 = \mathbf{s}_j \in \mathcal{S}, \\ \mathbf{B}_{\mathbf{v}_1} \tilde{\mathbf{C}}_{N(\mathbf{v}_1), N(\mathbf{v}_2)} \mathbf{B}_{\mathbf{v}_2}^\top + \mathbb{I}(\mathbf{v}_1 = \mathbf{v}_2) \mathbf{F}_{\mathbf{v}_1}, & \mathbf{v}_1, \mathbf{v}_2 \notin \mathcal{S}. \end{cases} \quad (36)$$

The three cases in (36) have straightforward interpretations. When both locations belong to $\mathcal{S}$, their covariance is obtained from the covariance structure already defined over the reference set. When only one location lies outside $\mathcal{S}$, its dependence with a reference location is transmitted through its nearest neighbors. When both locations are outside $\mathcal{S}$, their dependence is transmitted through the dependence between their respective neighbor sets. Finally, if $\mathbf{v}_1 = \mathbf{v}_2$, the term $\mathbf{F}_{\mathbf{v}_1}$ is added because the conditional residual uncertainty at that location also contributes to its variance. This construction is what allows the nearest-neighbor approximation to define a valid Gaussian process rather than merely an approximation to a finite-dimensional likelihood.

4.2 Choice of Reference Set and Neighbor Size

An important practical choice in implementing an NNGP is the specification of the reference set $\mathcal{S}$ and the number of neighbors m . A simple and effective choice is to take the reference set to be the set of observed locations, that is, $\mathcal{S} = \mathcal{T}$, where $\mathcal{T}$ denotes the locations at which observations are available.

The neighbor size m is generally chosen to be much smaller than the total number of locations. In practice, values in between 10 to 15 have often been found to provide performance close to that of the full GP for many spatial applications (Datta et al., 2016).

4.3 Computational Complexity and Performance

The computational advantage of NNGP follows from the fact that it never requires the inversion or factorization of the full dense covariance matrix. Instead, the construction repeatedly operates on covariance matrices involving at most m neighbors. For a reference set of size k together with n locations outside the reference set, the corresponding computational burden is of the order $O((n+k)m^3)$. Since m is typically small and treated as fixed, computation grows approximately linearly with the number of spatial locations, in contrast to the $O(N^3)$ growth of a full GP. The sparse precision structure also leads to substantial reductions in memory requirements. Empirical studies have shown that relatively small neighbor sets can provide inference and prediction comparable to those obtained from the parent Gaussian process, making NNGP particularly attractive for large spatial datasets (Datta et al., 2016; Finley et al., 2022).

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